

IntelliQuery: Official Technical Documentation

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1. System Architecture

IntelliQuery is an AI-driven Business Intelligence (BI) platform that utilizes a dual-engine approach to data analysis: the **SQL Planner** for structured data and the **RAG Agent** for unstructured knowledge.

Key Term: AI Planner

The system breaks down natural language queries into logical sub-tasks before executing code. This prevents "hallucinated" column names during DAX generation.

2. Data Handling & Limits

2.1 Upload Constraints

- **File Types:** Only .csv, .xlsx, and .pdf files are supported for local ingestion.
- **Size Limit:** Individual file uploads are restricted to **250 MB**.
- **Row Limit:** For free-tier workspaces, datasets are capped at 1,000,000 rows per model.

2.2 Connectivity

- **DirectQuery:** Available for PostgreSQL and MySQL only.
- **Refresh Intervals:** Scheduled refreshes occur every 12 hours by default.

3. The "Semantic Layer" Rules

To ensure the LLM generates accurate DAX measures, users must adhere to the following naming conventions:

- All measures must start with a prefix: **m_** (e.g., m_Total_Revenue).
- Column names must not contain special characters except underscores.
- **Reserved Keywords:** Words like "SUM", "AVG", and "TOTAL" are reserved by the backend and cannot be used as standalone measure names.

4. Visualization & AI Features

- **Anomaly Detection:** Requires a continuous date hierarchy and a minimum of **12 data points** (months/weeks).
- **Narrative Summaries:** Generates a text-based insight for any chart with more than two dimensions.
- **Exporting:** Reports can be shared via a secure HTTPS link or exported as an interactive HTML file.

5. Security Protocol

Data security is handled via **Row-Level Security (RLS)**. Policies are defined in the workspace settings using standard SQL WHERE clauses. Unauthorized attempts to access the HR_Finance table will trigger a system lockout for 30 minutes.